M339D: April 10th, 2026.

UNIVERSITY OF TEXAS AT AUSTIN

Problem Set # 13

Mean and median of the log-normal stock prices.

Problem 13.1. The current price of a non-dividend-paying stock is \$80 per share. Under the risk-neutral probability measure, its mean rate of return is 12% and its volatility is 30%.

Let $R(0,t)$ denote the realized return of this stock over the time period $[0,t]$ for any $t > 0$. Calculate $\mathbb{E}^*[R(0,2)]$.

$$\mathbb{E}^*[R(0,2)] = (r - \frac{\sigma^2}{2}) \cdot T = (0.12 - \frac{0.09}{2}) \cdot 2 =$$

$r = 0.12$

$$= (0.12 - 0.045) \cdot 2 = 0.15$$

Problem 13.2. A stock is valued at \$75.00. The continuously compounded, risk-free interest rate is 10.0% and the standard deviation of annualized returns is 25.0%. If the stock is lognormally distributed, what is the expected stock price after 2 years under the risk-neutral probability measure?

$\rightarrow$: $\mathbb{E}^*[S(T)] = \cancel{X}\ S(0)e^{rT}$

$$= 75 \cdot e^{0.10 \cdot 2} = 75 \cdot e^{0.2} = \underline{91.61}$$

Problem 13.3. A non-dividend-paying stock is valued at \$55.00 per share. Its standard deviation of annualized returns is given to be 22.0%. The continuously compounded risk-free interest rate is 12%. If the stock price is modeled using the lognormal distribution (as discussed in class), what is the median of the stock price in 3 years under the risk-neutral probability measure?

$$S(0)\, e^{(r - \frac{\sigma^2}{2}) \cdot T}$$

$$55\, e^{(0.12 - \frac{(0.22)^2}{2})(3)} = \$\ 73.31$$

Problem 13.4. Assume that the stock price is modeled using the lognormal distribution. The stock pays no dividends. Under $\mathbb{P}^*$, the annual mean rate of return on the stock is given to be 12%. Also under $\mathbb{P}^*$, the median time$-t$ stock price is evaluated to be $S(0)e^{0.1t}$. What is the volatility parameter of this stock price?

$\longrightarrow$: $r = .12$

$$S(0)e^{(r-\frac{\sigma^2}{2})\cdot T} = S(0)e^{0.1t}$$

$$r - \frac{\sigma^2}{2} = 0.1$$

$$.12 - \frac{\sigma^2}{2} = 0.1 \Rightarrow \frac{\sigma^2}{2} = 0.02$$

$$\sqrt{\sigma^2} = \sqrt{0.04} \quad \sigma = 0.2$$

Problem 13.5. The current stock price is \$100 per share. The stock price at any time $t > 0$ is modeled using the lognormal distribution. Assume that the continuously compounded risk-free interest rate equals 8%. Let its volatility equal 20%.

Find the value t^* at which the median stock price equals \$120, under the risk-neutral probability measure.

$\longrightarrow$: $S(0) = 100$, $r = 0.08$, $\sigma = 0.2$

$$S(0)e^{(r-\frac{\sigma^2}{2})t^*} = 120$$

$$100e^{(0.08-\frac{(0.2)^2}{2})t^*} = 120$$

$$(0.08 - \frac{(0.04)}{2})t^* = \ln(1.2)$$

$$0.06t^* = \ln(1.2)$$

$$t^* \approx 3.039$$

Problem 13.6. The volatility of the price of a non-dividend-paying stock is 0.2. The stock price is modeled using a log-normal distribution. Under $\mathbb{P}^*$, the expected time-2 stock price is \$120. What is the median of the time-2 stock price under $\mathbb{P}^*$?

$\longrightarrow$: $$\frac{\text{mean}}{\text{median}} = e^{\frac{\sigma^2}{2}\cdot T}$$

$$\text{mean } S(0)e^{rT} = 120$$

$$T = 2$$

$$\text{median} = \frac{\text{mean}}{e^{\frac{\sigma^2}{2}\cdot T}} = \frac{120}{e^{\frac{0.2^2}{2}\cdot 2}} = 115.295$$

Motivation.

Consider a European call option w/ strike K and exercise date T.

By our risk-neutral pricing

$$
\begin{aligned}
V_C(0) &= e^{-rT}\,\mathbb{E}^*[V_C(T)] \\
&= e^{-rT}\,\mathbb{E}^*[(S(T)-K)_+] \\
&= e^{-rT}\,\mathbb{E}^*[(S(T)-K)\cdot \mathbb{I}_{[S(T)\geq K]}] \\
&= e^{-rT}\,\boxed{\mathbb{E}^*[S(T)\cdot \mathbb{I}_{[S(T)\geq K]}]} - e^{-rT}\,\boxed{\mathbb{E}^*[K\cdot \mathbb{I}_{[S(T)\geq K]}]}
\end{aligned}
$$

$$\mathbb{E}[g(X)]$$

$$\vdots$$

$$S(0)e^{rT}\cdot N(d_1)$$

$$
\begin{aligned}
\boxed{\mathbb{E}^*[K\cdot \mathbb{I}_{[S(T)\geq K]}]} &= K\cdot \boxed{\mathbb{E}^*[\mathbb{I}_{[S(T)\geq K]}]} \\
&= \boxed{\mathbb{P}^*[S(T)\geq K]} \quad \checkmark
\end{aligned}
$$

w/ $d_1 = \frac{1}{\sigma\sqrt{T}}\left[\ln\left(\frac{S(0)}{K}\right) + \left(r+\frac{\sigma^2}{2}\right)\cdot T\right]$